Model Selection and Internal Validation

Applied Linear Models I • Lecture 12 of 16

Applied Linear Models I

The problem

Adding predictors never increases SSE – so we cannot select by fit alone.

- Penalize complexity: adjusted R^2, AIC, BIC, Mallows' C_p.
- Estimate prediction error directly: **cross-validation**.
- Beware the **optimism** of in-sample fit.

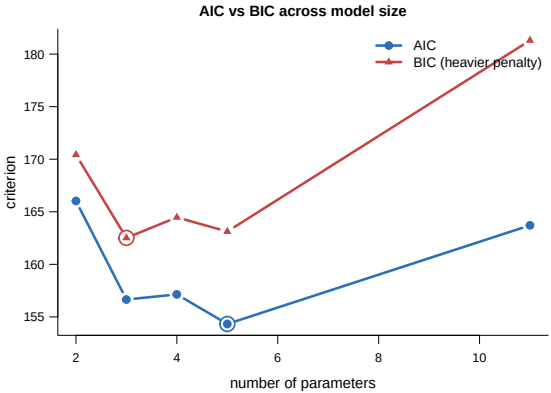
The criteria

Criterion	Formula (lower better, except adj R^2)	Penalty
adj R^2	$1 - \frac{\mathrm{SSE}/(n-p)}{\mathrm{SST}/(n-1)}$	light
AIC	$n \log(\mathrm{SSE}/n) + 2p$	$2p$
BIC	$n \log(\mathrm{SSE}/n) + p \log n$	$p \log n$ (heavier)
C_p	$\mathrm{SSE}_p/\hat{\sigma}^2_{\text{full}} - (n - 2p)$	good if $C_p \approx p$

On `mtcars`, the criteria need not agree on the winner.

AIC vs BIC

- AIC ($2p$): best **prediction**; keeps larger models.
- BIC ($p \log n$): identify the **true** model; consistent, more parsimonious.
- $C_p \approx$ AIC for linear models.



Stepwise selection – handle with care

Greedy add/drop to optimize a criterion. Cautions:

- explores a tiny, path-dependent slice of model space;
- **unstable** – small data changes flip the choice;
- post-selection p-values are not valid as prespecified.

One input, not a verdict. Pair with criteria, CV, and subject knowledge.

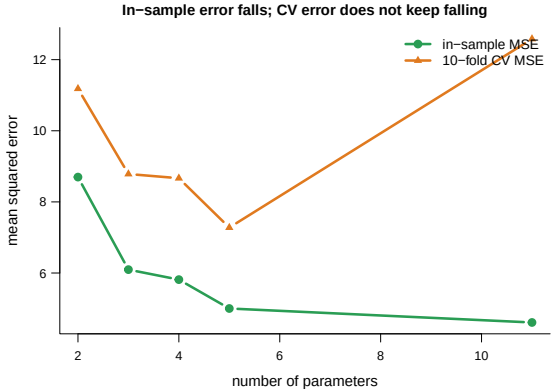
Cross-validation

Partition into K folds; fit on $K-1$, predict the held-out fold, average squared errors.

- Minimal assumptions; adapts to the data and loss.
- Cost: computation; variance from the fold split (repeat to reduce).
- Leave-one-out = PRESS via the leverage identity (Lecture 13).

The optimism of in-sample fit

Model	in-sample	CV
wt	8.70	11.18
wt+hp	6.10	8.78
wt+hp+qsec	5.81	8.67
wt+hp+qsec+am	5.00	**7.28**
full (10)	**4.61**	12.58



The full model fits best in-sample yet predicts worst – over-fitting. Never select by R^2.

Summary

- Penalize complexity: adj R^2 (light), AIC/C_p (prediction), BIC (parsimony).
- Stepwise is greedy and unstable – one input only.
- CV estimates prediction error directly.
- In-sample fit is optimistic; choose by CV, not R^2.
- All data-driven choices belong *inside* the CV loop.

Next

Lecture 13: external validation and PRESS.